

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. All working MUST be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.

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DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	$=$	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	$=$	9.80 m s^{-2}
1 Atmosphere	Atm	$=$	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	$=$	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water	ρ	$=$	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	C_w	$=$	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice	L_f	$=$	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water	L_v	$=$	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	$=$	$6.02 \times 10^{23} \text{ per mole}$
Molar gas constant	R	$=$	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	σ	$=$	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in free space	c	$=$	$3.00 \times 10^8 \text{ m s}^{-1}$
Planck's constant	h	$=$	$6.626 \times 10^{-34} \text{ Js}$

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) A ball of mass 1 kg is dropped from a height of h metres above the ground. The velocity, v , of the ball on its downward journey, is measured for various values of time, t . Table 1 shows the variation of velocity with time up to the point when the ball hits the ground.

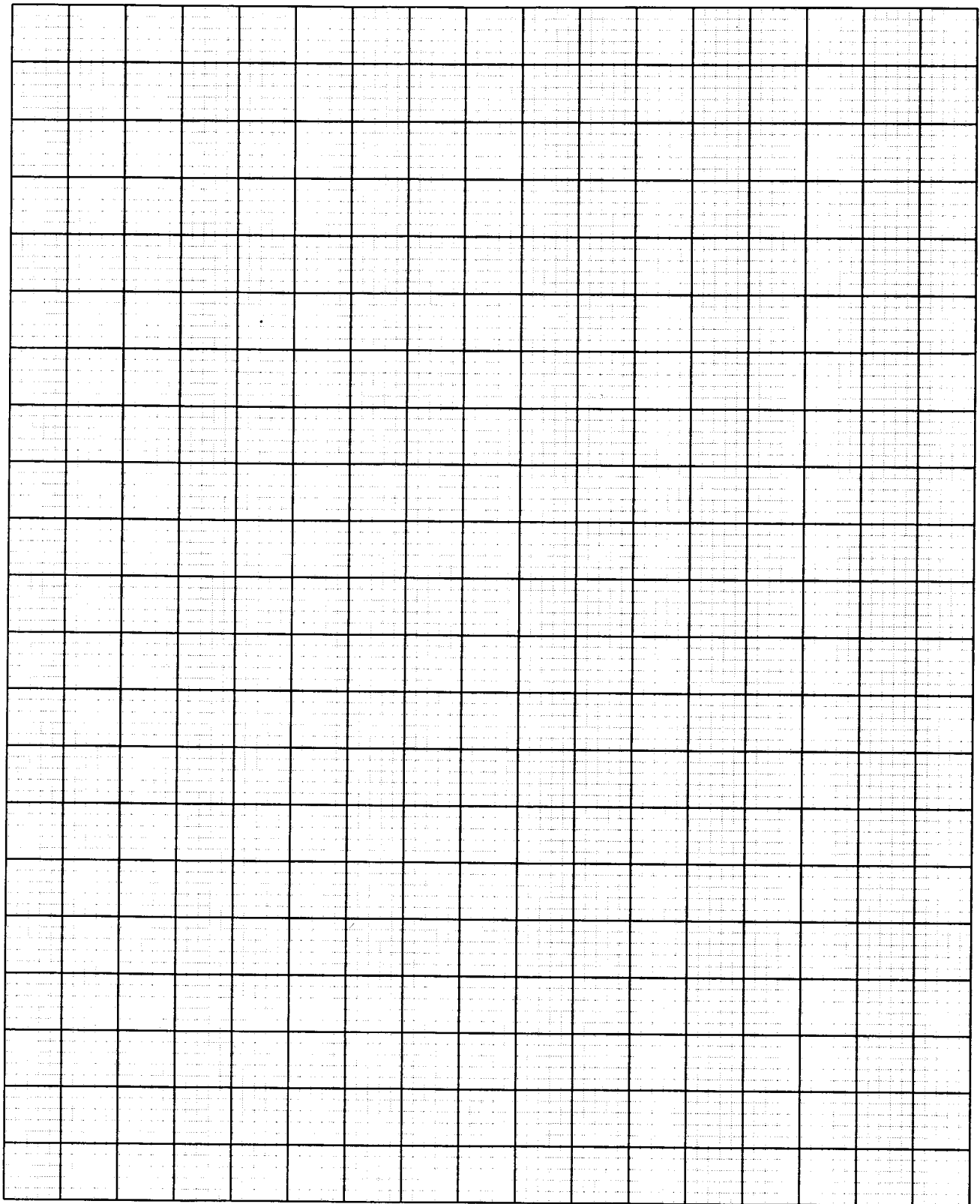
TABLE 1

Velocity $v/\text{m s}^{-1}$	Time t/s
0	0
	1.0
	2.0
	3.0
	4.0

- (i) Complete Table 1 by inserting the missing values. [1 mark]
- (ii) On the grid provided on page 5, plot a graph of velocity, v , against time, t . [4 marks]
- (iii) Using your graph, calculate the height, h , above the ground from which the ball was dropped.

[2 marks]

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- (b) A student standing on top of a cliff throws a stone with initial velocity, v_i , at an angle, θ , to the horizontal as shown in Figure 1.

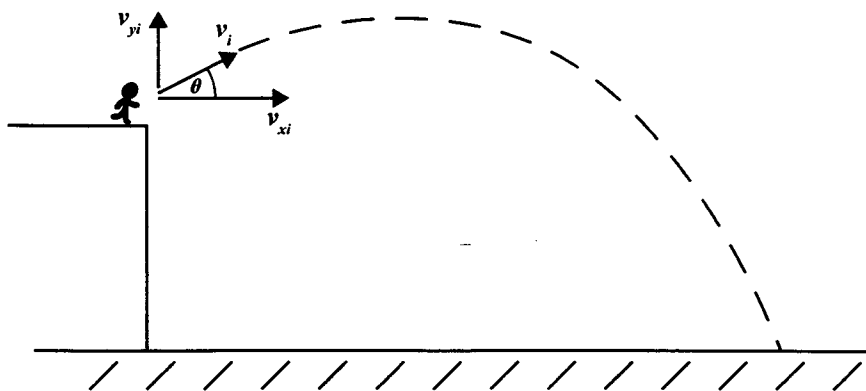


Figure 1

- (i) Write expressions for the initial vertical and horizontal components of velocity v_{yi} and v_{xi} respectively.

[2 marks]

- (ii) Write expressions for the vertical and horizontal displacements at any time t , for y and x respectively.

[2 marks]

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- (iii) Hence, show that the path traced out by the stone is parabolic.

[4 marks]

Total 15 marks

2. (a) Identify ONE similarity and ONE difference between transverse and longitudinal waves. Give an example of EACH type of wave.

[4 marks]

- (b) A rifle was fired in a valley. The echo was heard 8.3 seconds later. If the temperature in the valley is 10 °C, determine the distance between the shooter and the reflecting valley wall.

(Assume that the speed of sound = 340 m s⁻¹ at 0 °C and that for every 1 °C rise in temperature, the velocity of sound increases by 0.61 m s⁻¹.)

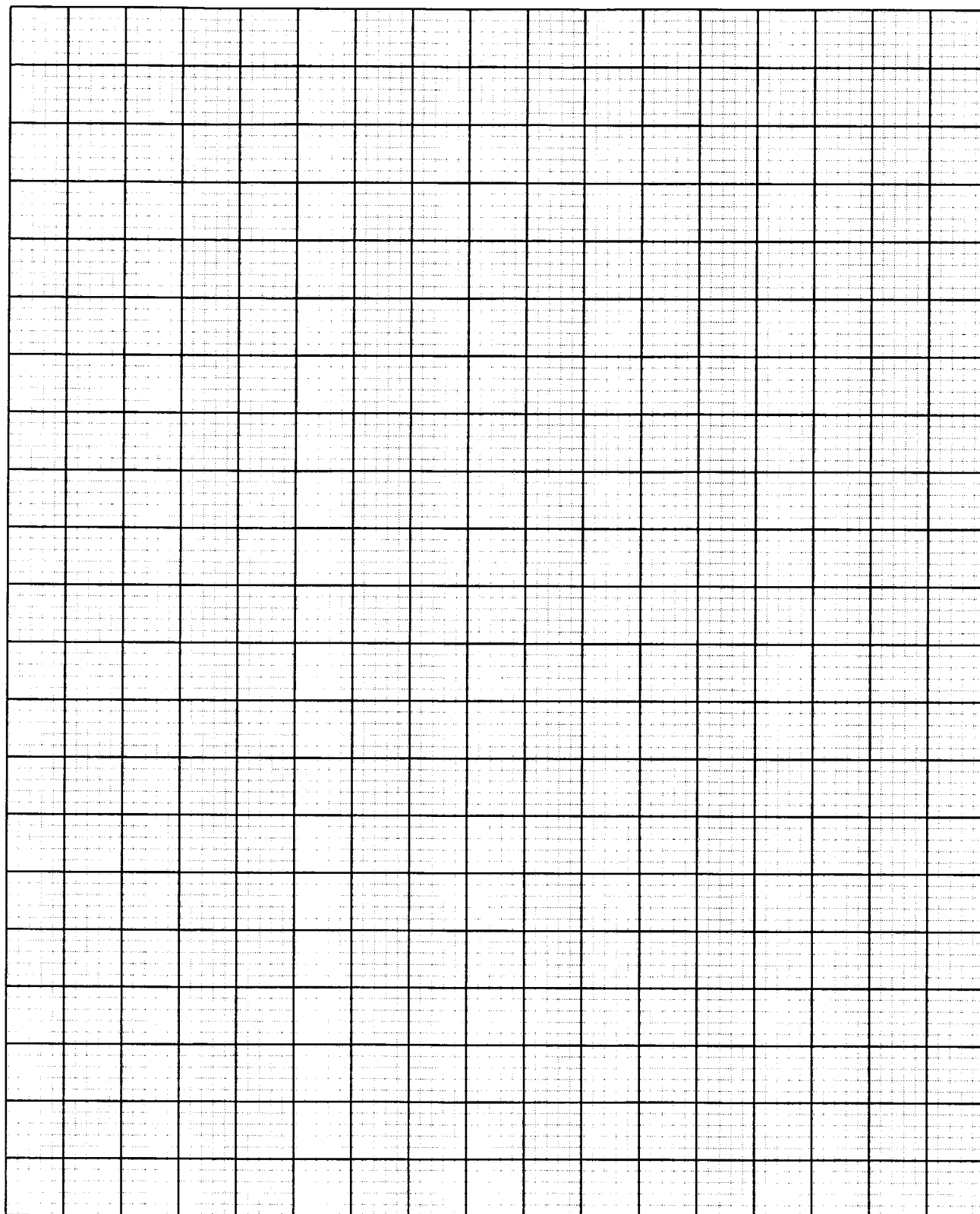
[3 marks]

- (c) Two students conduct an experiment to measure the speed of sound. The first student stands d metres away from a wall in a large empty school hall and claps his hands so that his claps coincide with the echoes from the wall. The second student measures the time for 20 claps [that is, the time from the first hand clap to the last echo after the twentieth hand clap]. The procedure is repeated for several values of d and the results are recorded in Table 2.

TABLE 2

d metres	t_{20} seconds	$t = \frac{t_{20}}{20}$ seconds	$\frac{t}{2}$ seconds
60	7.2		
70	8.2		
80	9.4		
90	10.6		
100	11.8		

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- (i) Complete Table 2 on page 8 by inserting the missing values. [2 marks]
- (ii) On the grid provided on page 9, plot a graph of d against $t/2$. Draw the best straight line through the points. [4 marks]
- (iii) Use your graph to determine the speed of sound, s .

[2 marks]

Total 15 marks

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3. (a) Show that both sides of the relation $PV = \frac{1}{3} N m \bar{c}^2$ have the same fundamental units.

[2 marks]

- (b) Using the equation in (a) above, show that the mean kinetic energy of the molecules in 1 mole of an ideal monatomic gas is equal to $(3/2) kT$, where k is the Boltzmann constant.

[3 marks]

- (c) Given that the root-mean-square (rms) speed of the molecules in a gas is inversely proportional to the square root of its molar mass, show that the ratio of the rms speeds of nitrogen to oxygen molecules in air is 1.07.

[Assume that the relative molar masses of nitrogen and oxygen are 28 and 32 respectively].

[2 marks]

- (d) 1.12 moles of gas in a closed container at a temperature of 273 K exert a pressure of $1.01 \times 10^5 \text{ N m}^{-2}$ on the walls of the container. Air is then extracted from the container so that the temperature is reduced to 223 K and the pressure reduced to $1.33 \times 10^4 \text{ N m}^{-2}$. Determine the number of moles in the container under these new conditions.

[3 marks]

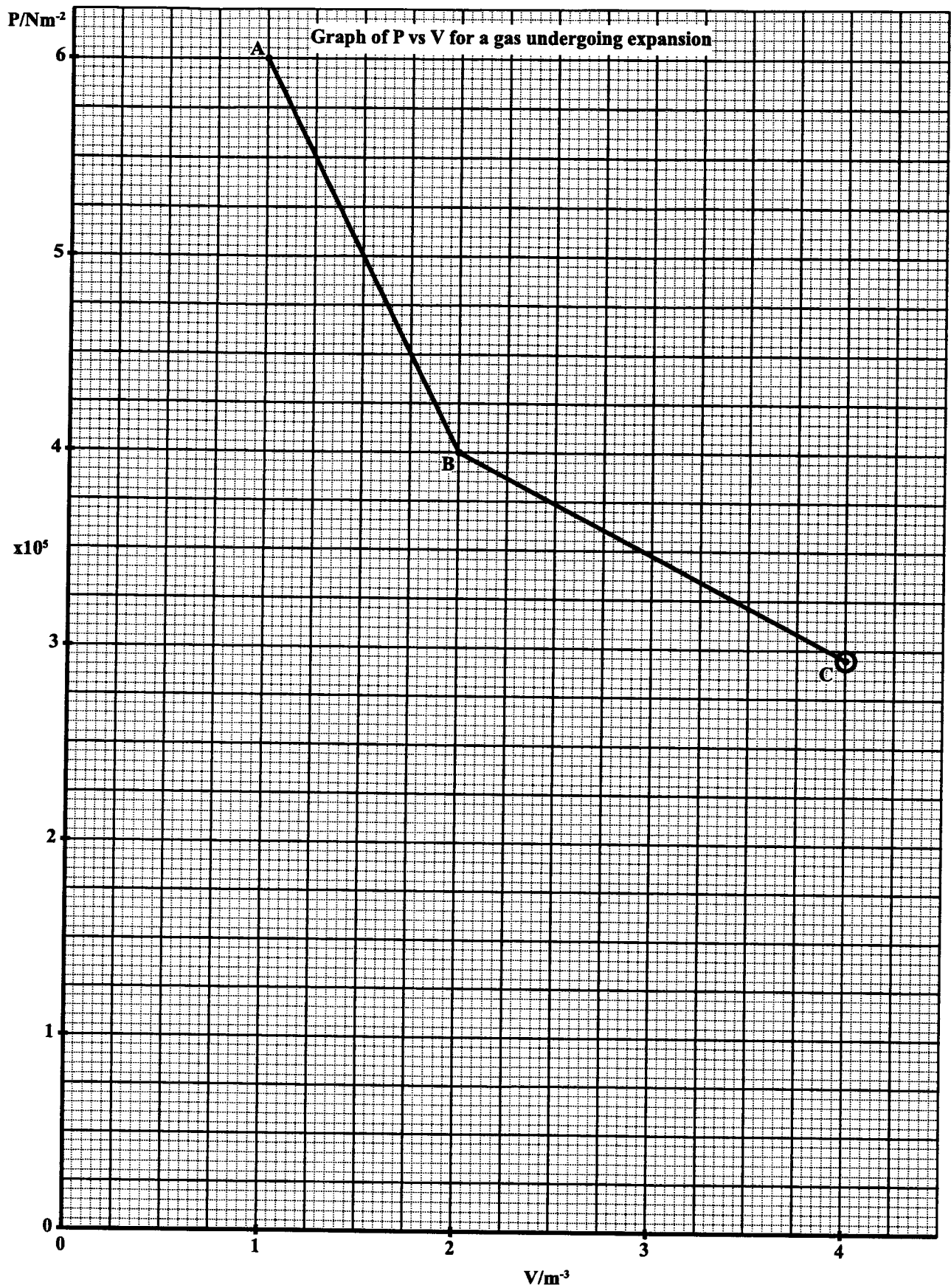
- (e) A gas expands along the path of ABC as shown on page 13. Use the graph to determine the work done in this expansion.

[3 marks]

- (f) If the expansion along AB had continued with no change in direction, what would be the volume of the gas at a pressure of $2 \times 10^5 \text{ N m}^{-2}$?

[2 marks]

Total 15 marks



SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. (a) State fully what is meant by the expression 'the Principle of Moments'. With the aid of an appropriate diagram, use the Principle of Moments to explain how it is possible for a person of mass 50 kg to lift someone twice as heavy on a see-saw. [5 marks]

- (b) (i) State the law of conservation of momentum.

Two objects, of masses m_1 and m_2 respectively, travelling in opposite directions collide. The respective velocities of the objects before the collision (u_1 and u_2) and after (v_1 and v_2) are given in the Table 3 below.

TABLE 3

Velocity before Collision	Velocity after Collision
$u_1 = +4 \text{ m s}^{-1}$	$v_1 = -1.5 \text{ m s}^{-1}$
$u_2 = -3 \text{ m s}^{-1}$	$v_2 = +5.5 \text{ m s}^{-1}$

- (ii) Assuming that the Law of Conservation of Momentum applies for this collision:
- Calculate the ratio of the masses of the two objects, $m_1:m_2$.
 - Calculate the total kinetic energy of the two objects before the collision, given that $m_1 = 8.5 \text{ kg}$.
 - Was this an elastic collision? Explain why or why not. [10 marks]

Total 15 marks

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Write the answer to Question 4 here.

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5. Figure 2 shows a stationary wave on a wire under tension. The frequency of the oscillations is 1200 Hz.



Figure 2

- (a) If the length of the wire, L , is 0.4 m, calculate the speed of the waves moving along it. [3 marks]
- (b) What will happen to the speed of the waves moving along the wire if
- (i) the tension is doubled (all other factors remaining the same)?
 - (ii) the length is doubled (all other factors remaining the same)? [2 marks]
- (c) The general equation for a travelling wave is $y = A_0 \sin \left(\frac{2\pi}{\lambda} x + 2\pi ft \right)$ where A_0 is the amplitude, λ is the wavelength and f is the frequency of the wave.

A sound wave is represented by the following equation:

$$y = 5 \times 10^{-4} \sin (740x + 251300t),$$

where x and y are measured in metres and t is measured in seconds.

Determine the following properties of the wave:

- (i) Amplitude
 - (ii) Wavelength
 - (iii) Frequency
 - (iv) Speed [7 marks]
- (d) Briefly describe a practical application in which this type of sound wave may be used. [3 marks]

Total 15 marks

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6. (a) With the aid of a diagram, derive the expression $P = h \rho g$ at a depth, h , in a liquid of density, ρ . [5 marks]
- (b) An aluminum cube, each side measuring 1 m, at sea level, lies on the sea floor at a depth of 2.5 km.

Determine the

- (i) pressure exerted on the cube at this depth

[Assume that the density of sea water is constant and equal to $1.04 \times 10^3 \text{ kgm}^{-3}$]

- (ii) decrease in length (in mm) of any side of the cube resulting from this pressure, given that the Young's Modulus for aluminum is 69 GPa. [5 marks]

- (c) The graph in Figure 3 shows the variation between force and extension for a thin piece of wire.

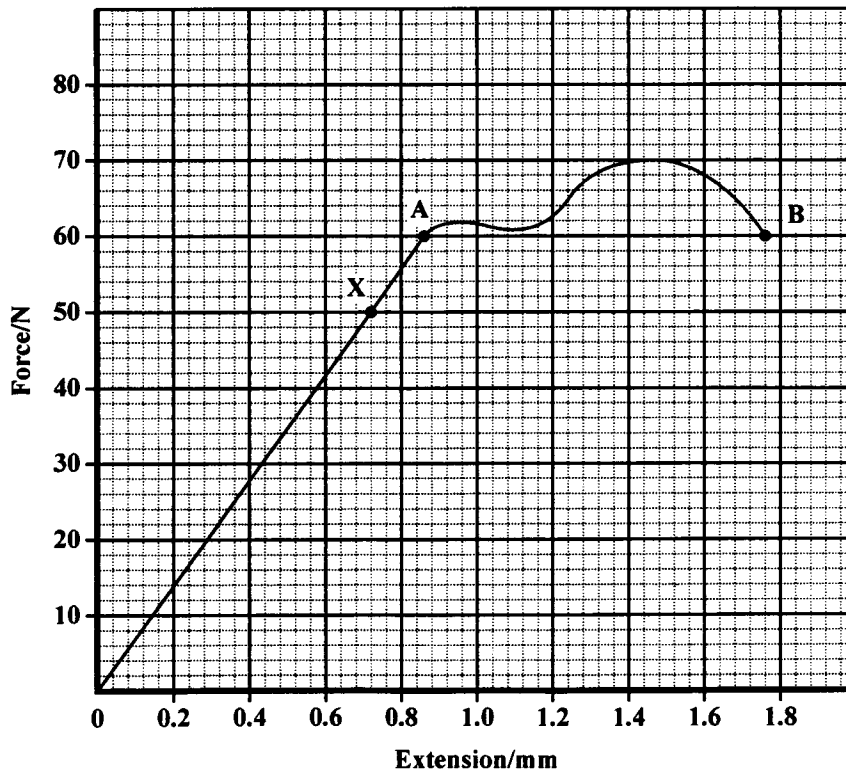


Figure 3

- (i) Explain what happens to the wire at points A and B.
- (ii) Using the graph, determine the strain energy in the wire at point X.

[5 marks]

Total 15 marks

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Write the answer to Question 6 here.

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IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.